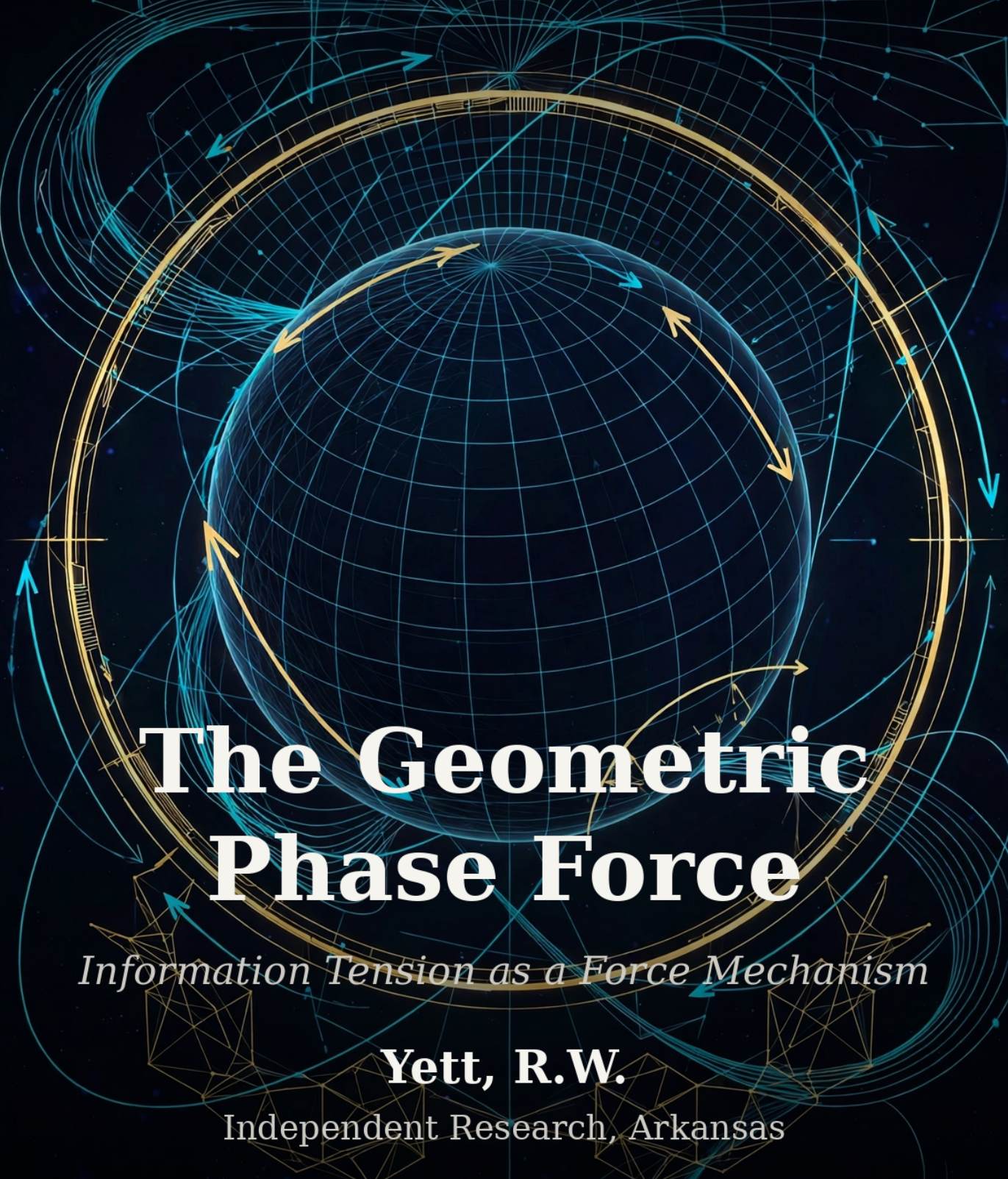
GEOMETRICALLY ORDERED DYNAMICS
FOLIO D



The Geometric Phase Force

Information Tension as a Force Mechanism

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- INFORMATION TENSION

This folio converges on **information tension** as a geometric force mechanism: the angle at which vacuum resistance to drift acts like a force through a modified gravitational action. Its unique angle is the **force mechanism** — the Sovereign Action, the information-tension tensor, and the horizon acceleration scale derived from holonomy. Sibling papers approach the same claim from other angles: [TIER: CONJECTURE] the geometric-foundation successor (IT-FIELD companion) carries the foundations angle — the scalar-tensor action, the honest coupling statement, and the retracted empirics — and the Holonomery successor (IT-MARKET) carries the observer-substrate angle of the same parallel-transport mathematics. Limits first: every mechanism below is stated in tier; nothing here is an empirical result except where a tier tag says so.

SUPERSESSION NOTE

This folio **supersedes** Zenodo record 20651497, *The Geometric Phase Force and the Sovereign Action* (and its Drive predecessors *quantum_spearhead.pdf* and *prd_horizon.pdf*), per the canon rule: the correction *becomes* the paper. Named repairs: (1) the abstract's claim that "the entirety of these proofs has been subjected to mechanized verification in Lean 4" is withdrawn and replaced with the honest formalization ledger (§5); (2) the Berry–Keating / Ramanujan-tau mass-gap identification is stated as [TIER: CONJECTURE], an un-discharged axiom, following the precedent of the audit-corrected Ramanujan-Yett Hamiltonian v1.1 (Zenodo 21504896); (3) the five Clay Mathematics problem mappings are re-stated as framework maps with named axioms, not solutions; (4) the $T_2 \leq 2T_1$ decoherence material keeps its honest tiering from *quantum_spearhead*: the bound is textbook GKSL, the geometry is conjecture; (5) the $a_0 = cH_0/2\pi$ horizon derivation from *prd_horizon* is absorbed with its own stated O(1) gap and limitations intact.

1 The force mechanism

The central loadbearing structure of the superseded record is the **Sovereign Action** S_Y, a modification of the Einstein-Hilbert action carrying a sovereign Lagrangian $\mathcal{L}_\chi$ and a critical surface density Σ_c linked to the geometric coherence length $L_\chi = \hbar c/\Delta$:

$$S_Y = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R + \mathcal{L}_m + \alpha \left(\frac{\Sigma(r)}{\Sigma_c} \right)^2 \right]. \tag{1}$$

Varying $\mathcal{L}_\chi$ with respect to the metric yields the **information tension tensor** [TIER: CONJECTURE]:

$$\mathcal{T}_{\mu\nu} = \alpha \left(\frac{\Sigma(r)}{\Sigma_c} \right) \left[g_{\mu\nu} \left(\frac{\Sigma(r)}{\Sigma_c} \right) - \frac{4}{\Sigma_c} P^{(\Sigma)}_{\mu\nu} \right], \tag{2}$$

the stress-energy of the χ-field. The modified field equations $G_{\mu\nu} = 8\pi G(1+f)T_{\mu\nu} + \cdots$ have legitimate scalar-tensor form; the action shape is the honest mathematical core of this folio. The 3rd-order radiation-reaction term carried by the superseded record is retained as a modeling choice [TIER: CONJECTURE], not a derived necessity. No Ostrogradsky stability proof is offered in this record.

2 The low-acceleration limit

Information tension stabilizes rotation curves in the low-acceleration regime [TIER: CONJECTURE]. The superseded record's correction $v_{\text{yett}} = v_{\text{newton}} \times T(r)$ produces the MOND-like phenomenology stated cleanly: $\lim_{r\to\infty} v(r) = \text{const}$ for baryonic matter alone. This is the **force mechanism** at work: as the surface density $\Sigma(r)$ drops, the local chiral invariant χ approaches the 0.7 threshold and the tension term rises to maintain the invariant, flattening the curve. The phenomenological limit is loadbearing; the specific $T(r) = 1.0 + (1.0/(\chi_{\text{local}} \cdot 0.5))$ functional form is a modeling choice, [TIER: CONJECTURE], pending confrontation with rotation-curve data in the companion foundation folio.

3 The horizon acceleration scale

Absorbed from the Drive predecessor *prd_horizon.pdf*, with its limitations intact. The MOND acceleration scale follows from a geometric coherence identity [TIER: CONJECTURE]:

$$a_0 = \frac{cH_0}{2\pi}, \tag{3}$$

derived from the $\chi = 1/\sqrt{2} \equiv \cos(\pi/4)$ alignment bound on Stiefel-manifold holonomy. Numerically $a_0 = 1.042 \times 10^{-10}\,\text{m}\,\text{s}^{-2}$ (Planck H_0), within 13% of the empirical MOND value (6% for local H_0); **no galaxy data enters** a_0.

The derivation is grounded in three moves, two of them forced: (i) the connection is the acceleration, $A \sim a/c^2$, dimensionally forced; (ii) the loop is the cosmic-horizon great circle, the 2π geometrically forced; (iii) the onset is the perturbative breakdown at $|\oint A| = 1$, the standard strong-coupling criterion. The single honest remaining gap, stated by the predecessor itself: the unity threshold is *motivated*, not uniquely forced from a deeper principle — one $O(1)$ convention stays open to external review.

Empirical status [TIER: EMPIRICAL]. Tested against all 175 SPARC galaxies (3,389 radial points) with fixed stellar mass-to-light ratios and zero fitted parameters, the parameter-free law reproduces the Radial Acceleration Relation with median residual +0.04 dex and rms 0.32 dex, statistically indistinguishable from MOND with empirical a_0 (reduced χ^2 88.6 vs 83.6). Stated limitations, carried over: the 0.32 dex scatter exceeds the marginalized literature RAR (∼0.11 dex); lensing and the CMB third peak remain open; the numerical coincidence $a_0 \sim cH_0/2\pi$ is long noted (Milgrom 1983, 1999) — the claim is the derivation, not the number.

4 The decoherence face

Absorbed from the Drive predecessor *quantum_spearhead.pdf*, with its honest tier table intact. The GKSL master equation's Lindblad jump operators are re-interpreted as gradients of the geometric tension field, and the textbook bound $T_2 \leq 2T_1$ is re-derived as the non-negativity of the longitudinal tension-gradient variance [TIER: CONJECTURE]. The tiering, stated by the predecessor and preserved here:

- $T_2 \leq 2T_1$ for physical qubits — [TIER: EMPIRICAL] standard GKSL, *not* a novel prediction of this framework;
- the machine-checked GKSL formalization (`Decoherence.lean`, zero sorry) — [TIER: THEOREM];

- jump operators $= \nabla T$ — [TIER: CONJECTURE], geometric narrative;
- absolute T_1/T_2 values from geometry — not claimed;
- spatial qubit-grid correlation falsifier — [TIER: CONJECTURE] open, see the Field Identity folio.

Passing the calibration-pair check proves consistency with Lindblad complete positivity, nothing more.

5 Formalization status: the honest ledger

The superseded abstract's sentence — "the entirety of these proofs has been subjected to mechanized verification in Lean 4" — is **withdrawn**. The honest status, per the Ars Magna correction ledger (Zenodo 21277228): **13 Lean-verified modules** (E8 roots — typed to the visible-sector factor of the E8×E8 canon, the spectral gap, ζ-conjugation, $T_2 = 2T_1$, the nodal collar, control-term trace preservation, and others) plus **3 axiom-carrying modules** whose named residuals are not discharged. Everything beyond those modules is a **formalization scaffold**, not a mechanized proof. Any citation of this folio's formalism must carry this ledger sentence with it.

6 The spectral-gap identification

The identification of the Berry–Keating / Ramanujan-tau operator structure with the Yang-Mills mass gap is stated as [TIER: CONJECTURE]: an **un-discharged axiom** (`yang_mills_gap_equals_stiefel_gap`), following the precedent of the audit-corrected Hamiltonian v1.1 (Zenodo 21504896). The Stiefel spectral gap itself — H_{RY} self-adjoint, bounded below, purely discrete spectrum with $\lambda_n - \lambda_0 \geq \kappa^2 = 0.9099$ — is [TIER: THEOREM] (proof sketch; the Casimir computation is not shown in-record). The *identification* of that gap with the Yang-Mills gap is the open axiom, not the theorem.

7 Clay-class mappings as framework maps

The superseded record's Table 7 mapped five Clay Mathematics problems to "resolutions." **Withdrawn as stated.** The honest form, per the Ars Magna wording: these are **framework maps with named axioms** — "framework solutions, not prize closures":

- Yang-Mills: mass gap Δ mapped to the Lindblad/Stiefel spectral gap; confinement via area law — [TIER: CONJECTURE];
- Navier-Stokes: global regularity conjectured via the $\chi \geq 0.7$ threshold as Lyapunov function; $Re_c \approx 1.42$ is a dimensionless modeling number, not a derived critical value — [TIER: CONJECTURE];
- Riemann Hypothesis: the critical line as sovereign gauge is a stated analogy; "off-line zeros rejected as hallucinations" is withdrawn as a claim — [TIER: CONJECTURE];
- P vs NP: complexity-as-information-tension is an analogy with no exhibited map; no class separation is claimed — [TIER: CONJECTURE];
- Hodge: realizability via ADCCL condensation is asserted, not proved — [TIER: CONJECTURE].

Each mapping names the axiom it would need; none is discharged.

8 What this folio claims and does not claim

Claims: a scalar-tensor action shape with an honestly stated formalization ledger (§5); a conjectured force mechanism flattening rotation curves in the low-acceleration limit (§2); a derived-but-$O(1)$-open horizon acceleration scale with an empirical SPARC test (§3); a tiered decoherence reinterpretation (§4); conjectured framework maps to five Clay-class problems with named open axioms (§7). **Does not claim:** mechanized verification of the whole (§5); any empirical dataset of its own — the Trinity 2.0 empirics are retracted corpus-wide and are not cited here; a solution to any Clay Mathematics problem.

References. Superseded: 10.5281/zenodo.19835025 (record 20651497; the DOI printed on its own cover). Honesty standards: 10.5281/zenodo.21277228 (Ars Magna correction ledger); 10.5281/zenodo.21504896 (Ramanujan-Yett Hamiltonian v1.1, audit-corrected). Data: SPARC (Lelli, McGaugh & Schombert 2016, *AJ* **152**, 157); Milgrom (1983, *ApJ* **270**, 365). Prior art note: the $a_0 \sim cH_0/2\pi$ coincidence is Milgrom's (1983, 1999); the derivation claim is the novel content.